

Packet Tracer – Advanced ACL and Network Security Policies

Objectives

Part 1: Build a Multi-Network Topology

Part 2: Configure IP Addressing

Part 3: Verify Basic Connectivity

Part 4: Configure a Standard ACL

Part 5: Configure an Extended ACL

Part 6: Configure Server Services

Part 7: Verify Security Policies

Background / Scenario

You are the network administrator for a small company with two departments:

- HR Department
- Sales Department

Management requires the following security policies:

- HR users can access all resources.
- Sales users cannot send ICMP traffic to the HR network.
- Sales users may access the company web server using HTTP.
- Sales users cannot access the FTP service.
- All other traffic should be permitted.

You will use both Standard and Extended Access Control Lists (ACLs) to implement these requirements and verify traffic filtering.

This lab demonstrates how ACLs provide traffic filtering and access control in enterprise networks.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 1 Router (2911)
- 2 Switches (2960)
- 1 Server
- 2 PCs

Rename:

- Router0 → R1
 - Switch0 → SW-HR
 - Switch1 → SW-SALES
 - Server0 → WEB-SERVER
 - PC0 → HR-PC
 - PC1 → SALES-PC
-

Step 2: Connect Devices

Use Copper Straight-Through cables.

Device	Interface	Device	Interface
HR-PC	Fa0	SW-HR	Fa0/1
WEB-SERVER	Fa0	SW-HR	Fa0/2
SW-HR	Fa0/24	R1	G0/0
SALES-PC	Fa0	SW-SALES	Fa0/1
SW-SALES	Fa0/24	R1	G0/1

Part 2: Configure IP Addressing

Network Plan

Device	IP Address	Mask	Gateway
HR-PC	192.168.10.10	255.255.255.0	192.168.10.1
WEB-SERVER	192.168.10.100	255.255.255.0	192.168.10.1
SALES-PC	192.168.20.10	255.255.255.0	192.168.20.1
R1 G0/0	192.168.10.1	255.255.255.0	N/A
R1 G0/1	192.168.20.1	255.255.255.0	N/A

Part 3: Configure Router Interfaces

On R1:

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

interface g0/1

ip address 192.168.20.1 255.255.255.0

no shutdown

end

Verify Connectivity

From SALES-PC:

ping 192.168.10.10

Expected:

Reply from 192.168.10.10

From HR-PC:

ping 192.168.20.10

Expected:

Reply from 192.168.20.10

From SALES-PC:

ping 192.168.10.100

Expected:

Reply from 192.168.10.100

At this stage all devices should communicate normally.

Part 4: Configure a Standard ACL

Security Requirement

Block SALES-PC from accessing the HR network.

Step 1: Create ACL

On R1:

configure terminal

access-list 10 deny host 192.168.20.10

access-list 10 permit any

Step 2: Apply ACL

Apply ACL near the destination network.

```
interface g0/0
```

```
ip access-group 10 out
```

```
end
```

Verify Standard ACL

Test 1

From SALES-PC:

```
ping 192.168.10.10
```

Expected:

Request timed out

Test 2

From SALES-PC:

```
ping 192.168.10.100
```

Expected:

Request timed out

Test 3

From HR-PC:

```
ping 192.168.20.10
```

Expected:

Request timed out

Why?

Standard ACLs only examine the source IP address.

The return ICMP reply from SALES-PC has source IP 192.168.20.10, which is denied by ACL 10.

This behavior is expected.

Part 5: Remove Standard ACL

Before continuing:

configure terminal

interface g0/0

no ip access-group 10 out

exit

no access-list 10

end

Verify:

show ip interface g0/0

Expected:

Inbound access list is not set

Outbound access list is not set

Part 6: Configure Extended ACL

Security Policy

Allow:

- HTTP from Sales to WEB-SERVER

Block:

- ICMP from Sales to HR Network
- FTP from Sales to WEB-SERVER

Permit:

- All other traffic

Step 1: Create Extended ACL

configure terminal

ip access-list extended SECURITY-POLICY

deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255

deny tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 21

permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 80

permit ip any any

exit

Step 2: Apply ACL

Apply near the source.

interface g0/1

ip access-group SECURITY-POLICY in

end

Part 7: Configure Services

Configure HTTP Service

On WEB-SERVER:

Services → HTTP

Enable:

HTTP = ON

Configure FTP Service

On WEB-SERVER:

Services → FTP

Enable:

FTP = ON

Create:

Username: admin

Password: cisco

Verify Server Configuration

WEB-SERVER:

IP Address: 192.168.10.100

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Part 8: Verify Security Policies

Test 1: ICMP Block

From SALES-PC:

ping 192.168.10.100

Expected:

Request timed out

Test 2: HTTP Allowed

From SALES-PC:

Desktop → Web Browser

Browse:

http://192.168.10.100

Expected:

Web page loads successfully

Test 3: FTP Block

From SALES-PC:

ftp 192.168.10.100

Expected:

Connection failed

Test 4: HR Access

From HR-PC:

ping 192.168.20.10

Expected:

Request timed out

Verify ACL Counters

On R1:

show access-lists

Expected:

Extended IP access list SECURITY_POLICY

deny icmp

(matches increasing)

deny tcp eq 21

(matches increasing)

permit tcp eq 80

(matches increasing)

Verify Interface ACL

show ip interface g0/1

Expected:

Inbound access list is SECURITY_POLICY

Troubleshooting Tips

If filtering does not work:

✓ Verify ACL is applied to the correct interface

show run

✓ Verify ACL direction

in

out

✓ Verify wildcard masks

0.0.0.255

✓ Verify ACL order

ACLs process top-to-bottom.

✓ Verify Standard ACL was removed before testing Extended ACL.

✓ Verify WEB-SERVER default gateway is configured.

✓ Verify HTTP and FTP services are enabled.

✓ Use Simulation Mode to observe packet filtering.

What You Learned

- Difference between Standard and Extended ACLs
- How ACLs filter traffic
- How ACLs inspect source IP addresses
- How Extended ACLs inspect protocols and ports
- How to block ICMP traffic
- How to block FTP traffic
- How to permit HTTP traffic
- How to verify ACL operation using counters

Challenge Extensions (Optional)

Challenge 1: Block HTTPS

Add:

```
deny tcp any any eq 443
```

Test HTTPS access.

Challenge 2: Create Guest Network

Add:

192.168.30.0/24

Allow Internet access only.

Block access to internal networks.

Challenge 3: Time-Based ACL

Research Cisco Time-Based ACLs.

Permit access only during business hours.

Challenge 4: Protect Router Management

Permit SSH only from HR-PC.

```
permit tcp host 192.168.10.10 any eq 22
```

```
deny tcp any any eq 22
```

Challenge 5: Simulation Mode Analysis

Switch Packet Tracer to:

Simulation Mode

Generate:

- ICMP traffic
- HTTP traffic
- FTP traffic

Observe exactly where the ACL permits or drops packets.